Data Analyst Intern: Bindiya Anand (Inter563)

IndiaKart E-Commerce - Data Quality Report

Phase 1: Data Cleaning & Quality Check

Kindly view the Python Notebook file in this link: Python Notebook file link

1. Objective

Phase 1 assessed the completeness, consistency, validity, uniqueness and referential integrity of the eight IndiaKart datasets (Orders, Order Items, Customers, Products, Payments, Returns, Inventory, Suppliers) before further analysis.

2. Dataset Overview

Table 1 shows the record and attribute counts of each dataset as loaded. Order Items is the largest table (100,000 rows); Suppliers is the smallest (200 rows).

Table 1: Dataset overview

Dataset	Records	Attributes
Orders	6,543	19
Order Items	100,000	11
Customers	10,000	17
Products	1,000	17
Payments	18,930	13
Returns	10,000	10
Inventory	1,000	11
Suppliers	200	14

3. Missing Value Assessment

Table 2 lists every column with missing values. Only Orders and Payments contain any; Order Items, Customers, Products, Returns, Inventory and Suppliers are fully populated.

Table 2: Missing value summary

Dataset	Column	Missing Count	Missing %
Orders	delivered_date	2,257	34.49
Orders	is_cod	1	0.02
Orders	channel	1	0.02

Dataset	Column	Missing Count	Missing %
Payments	order_id	1	0.01
Payments	customer_id	1	0.01
Payments	payment_date	1	0.01
Payments	payment_time	1	0.01
Payments	payment_method	1	0.01
Payments	amount	1	0.01
Payments	status	1	0.01
Payments	transaction_id	1	0.01
Payments	bank_name	1	0.01
Payments	gateway	1	0.01
Payments	refund_amount	1	0.01
Payments	refund_date	18,930	100.00

- **Expected / explainable:** delivered_date (2,257 orders, 34.49%) is blank for orders without a delivery date; this is consistent with orders not yet delivered, cancelled or otherwise not completed. The 100% blank Payments refund_date is consistent with refund_amount being 0.0 on all populated payment records.
- **Genuine concern:** one Payments record (PAY018930) is blank in every field except payment_id, i.e. an incomplete record. One Orders record is also missing is_cod and channel; the notebook does not investigate this further.

4. Data Quality Findings

Table 3 consolidates the checks run across the datasets and their outcome.

Table 3: Final data quality findings

Dataset	Check	Result	Unit	Status
Orders	Duplicate order_id	0.00	duplicates	Pass
Orders	Missing delivered_date	2257.00	records	Reviewed
Orders	Invalid delivered_date	0.00	records	Pass
Orders	final_amount > ₹5 lakh	52.00	orders	Reviewed
Orders	Tracking ID linked to multiple orders	0.00	tracking IDs	Pass
Order Items	Duplicate order_id + product_id combinations	103.00	combinations	Reviewed

Dataset	Check	Result	Unit	Status
Order Items	Records involved in repeated combinations	206.00	records	Reviewed
Order Items	Order IDs missing from Orders	83612.00	records	Issue
Customers	Customers with total_orders = 0	457.00	customers	Not an issue
Payments	Successful payments	18252.00	payments	Informational
Payments	Failed payments	677.00	payments	Reviewed
Payments	Payment failure rate	3.58	%	Reviewed
Payments	Payment records with invalid order IDs	12387.00	records	Issue
Returns	Returns without matching order ID	8022.00	records	Reviewed
Inventory	Out of Stock products	2.00	products	Reviewed

- **Uniqueness (Orders):** no duplicate order_id and no tracking_id linked to more than one order. Orders can be safely used as a unique-key table.
- **Delivery-date validation:** all 4,286 populated delivered_date values fall after the order_date (0 invalid). Delivery-time metrics are therefore logically consistent, but must be calculated on delivered records only.
- **High-value orders:** 52 orders exceed ₹5 lakh (maximum ₹9.43 lakh). They were flagged, not removed; they can skew averages and should be handled deliberately in KPI work.
- **Order Items:** 103 order_id + product_id combinations occur twice (206 records). In the reviewed records the repeated lines differ in quantity and amounts, so they may be separate line items rather than errors; revenue and item counts should be aggregated with this in mind.
- **Referential integrity:** 83,612 order-item records, 12,387 payment records (including the blank record) and 8,022 return records reference an order_id absent from the Orders dataset. The Orders file holds only 6,543 orders, so joins to Orders will drop most dependent records unless the cause is resolved.
- **Customers:** 457 customers have total_orders = 0; these are registered customers with no recorded orders and are not treated as an error.
- **Payments:** 18,252 payments succeeded and 677 failed, a 3.58% failure rate, plus the one incomplete record noted above. Revenue-related analysis should use successful payments only.
- **Inventory:** 2 products are Out of Stock; all inventory product_ids were found in Products (0 unmatched).

5. Cleaning / Treatment Summary

- Converted date columns to datetime: order_date, delivered_date, registration_date, last_login_date, launch_date, payment_date, return_date, refund_date, last_restocked_date, created_date.
- Parsed order_time and payment_time into time objects.
- Identified and profiled missing values, duplicate keys and repeated order/product combinations.
- Validated relationships between datasets (Order Items, Payments, Returns to Orders; Inventory to Products) and delivery-date logic.
- Flagged anomalies (orders above ₹5 lakh, incomplete payment record). **No records were deleted, imputed or corrected in the notebook.**

6. Overall Data Quality Assessment

The datasets are suitable for further analysis with documented data-quality considerations. Key fields are unique, data types are now analysis-ready and delivery dates are logically valid. The main limitation is the large referential-integrity gap against Orders. Points to keep in mind during EDA, SQL, KPI and Power BI work:

- Resolve or document the Orders coverage gap before joining tables; otherwise results will be based on a small share of Order Items, Payments and Returns.
- Treat blank delivered_date as meaningful; compute delivery KPIs on delivered orders only.
- Exclude failed payments from revenue; exclude or review the incomplete payment record.
- Decide how to treat the 52 high-value orders and the 103 repeated order/product combinations.